Predicting Unmet Healthcare Needs Among Adults with Disabilities

A Survey-Weighted Machine Learning Approach

Guang-Hua Liao, Coto Nagamine, Julian Newman • Wichita State University

Introduction

Adults with functional limitations face stacked barriers to care — cost, distance, provider availability, and unequal resource allocation — that accumulate as unmet need: forgone care, missed checkups, prolonged poor health. BRFSS measures these indicators annually, but descriptively.

We train a survey-weighted classifier on the full pooled BRFSS cohort and evaluate its targeting performance on adults with 1+ functional limitations — the subpopulation in which unmet need most concentrates. We then audit calibration slopes across race, household income, and limitation subtype.

The question: does risk mean the same thing across these dimensions?

Data Source & Preparation

- **Data:** Behavioral Risk Factor Surveillance System (CDC), a random-digit-dial survey of US adults in all 50 states and DC. Waves 2016–2024 pooled.
- **Split:** 80/20 random train/test on the full pool (3,968,572 adults) → 3,174,858 training / 793,714 test. Evaluation restricted to the 233,721 test respondents with 1+ functional limitations.
- **Outcome:** Binary union of six BRFSS items — cost-related forgone care, uninsured, no personal provider, no recent checkup, ≥14 poor mental-health days, ≥14 poor physical-health days.
- **Features:** 30+ demographic, socioeconomic, clinical, and behavioral variables, plus six binary limitation-subtype flags (hearing, vision, mobility, cognitive, independent-living, self-care). Special codes (DK/Refused/None) harmonized across waves.
- **Weights:** Complex survey weights applied at every stage — imputation, correlation screening, weighted gradient updates, calibration audit.

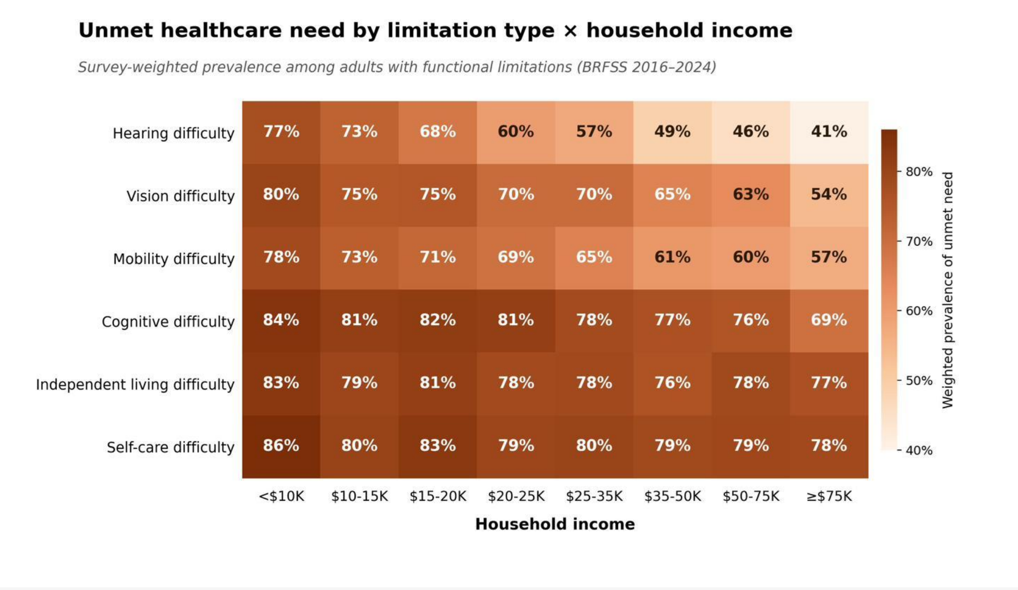

Figure 1. Survey-weighted prevalence of unmet healthcare need among adults with functional limitations, stratified by limitation type and household income. BRFSS 2016–2024 pooled.

Unmet need varies by more than 40 points across the cohort. Hearing, vision, and mobility limitations show a steep income gradient; cognitive, independent-living, and self-care limitations sit at or above 75% across all income levels.

A single descriptive rate for "adults with disabilities" hides this structure — motivating individual-level prediction.

Results

Three survey-weighted classifiers trained on identical preprocessing: L2-regularized logistic regression (baseline), random forest, and CatBoost (primary). All metrics below evaluated with survey weights on the held-out test set.

Model	AUC	Brier	F1
Ridge Logistic	0.782	0.179	0.796
Random Forest	0.789	0.176	0.804
CatBoost	**0.806**	**0.168**	**0.816**

Held-out test performance on adults with ≥1 functional limitation (n = 233,721). All metrics computed with survey weights. CatBoost wins on all three.

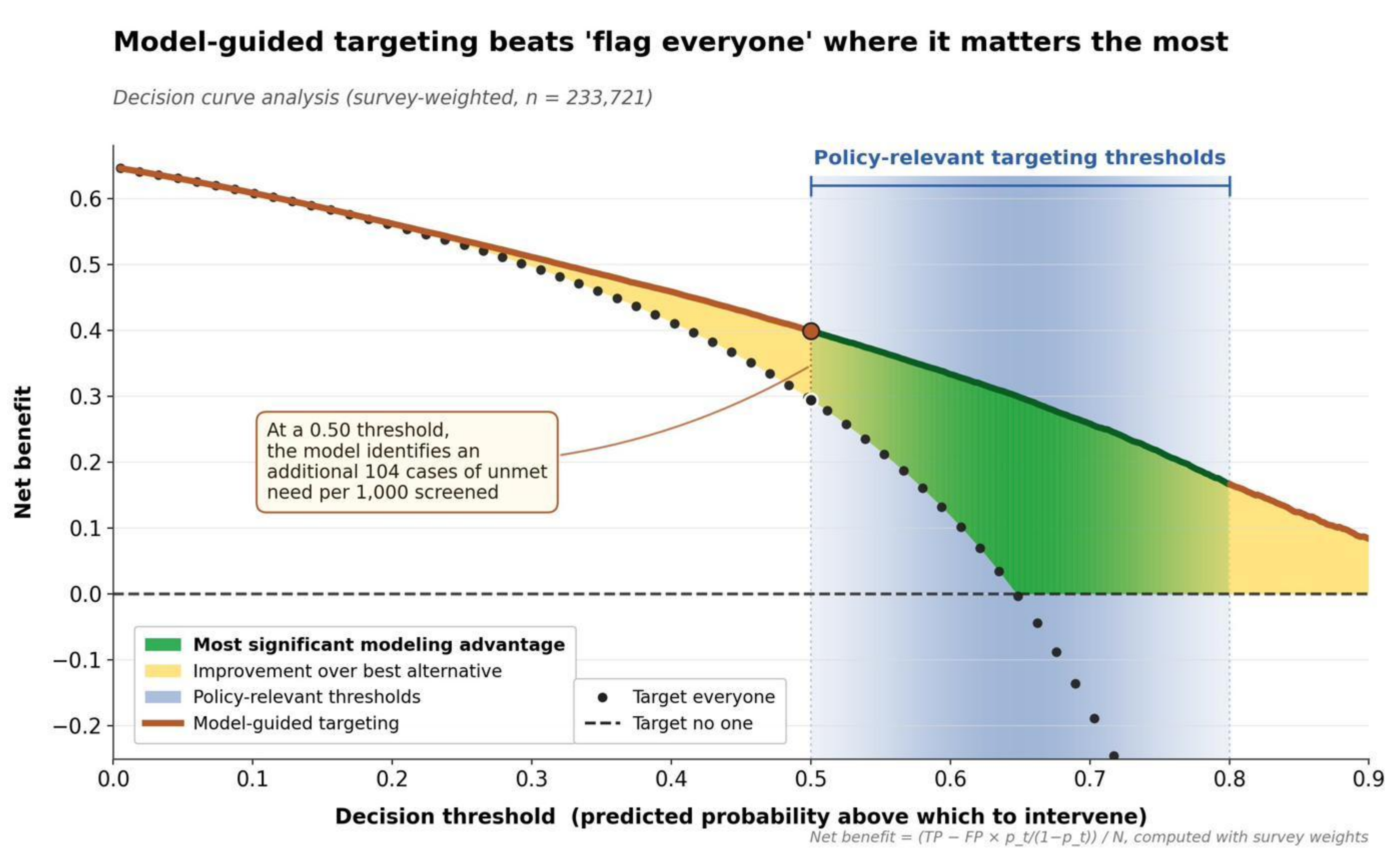

Figure 2. Survey-weighted decision curve analysis on the evaluation subpopulation (held-out test respondents with ≥1 functional limitation, n = 233,721). Shaded green: CatBoost's net-benefit gain over the better default strategy at each threshold. Blue band: policy-relevant 0.5–0.8 targeting range.

The gains in AUC and Brier are meaningful only if they change decisions. Decision curve analysis asks whether acting on the model produces more net benefit than the defaults — "flag everyone" or "flag no one" — across the threshold range a policymaker might consider.

Across the full policy-relevant 0.5–0.8 range, CatBoost-guided targeting dominates both defaults. At a 0.5 threshold, it identifies an additional 104 cases of unmet need per 1,000 screened over "flag everyone" — turning discrimination gains into real decision improvements.

Fairness & Calibration

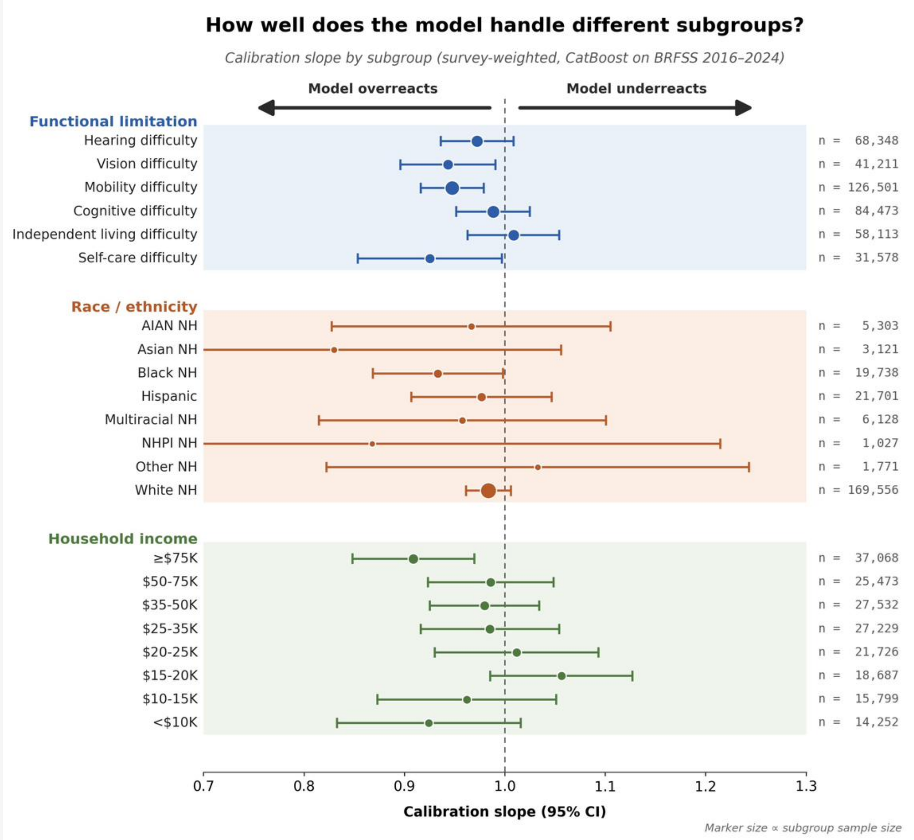

Figure 3. Survey-weighted calibration slopes with 95% CIs. Marker size ∝ subgroup n. Slope = 1 (dashed): ideal. < 1: model overreacts. > 1: underreacts.

Aggregate AUC and Brier mask a question every deployable model must answer: does it work equally well on the subgroups that matter?

Across 23 subgroups, calibration slopes fall within one CI of an ideal 1.0 — the model is appropriately calibrated on limitation types, all race/ethnicity groups, and 8 of 9 income brackets.

The exception is the ≥$75K bracket (slope ≈ 0.91), where the model systematically overreacts. This would have been invisible in an aggregate-only audit and warrants within-subgroup recalibration before deployment.

CatBoost's predictions rest on clinically intuitive drivers. Four features tell four different stories:

- **General health**: monotone dose-response (Poor ≫ Excellent).
- **Age**: smoothly decreasing — likely Medicare's cost buffer after 65.
- **Depression**: a large binary shift.
- **Sleep**: non-monotone U-shape; lowest risk at 7–9 hours — a pattern a linear baseline misses.

Figure 4. Survey-weighted SHAP dependence for CatBoost's four most influential features. Orange dots: individual respondents; green line: weighted population average.

Discussion

Targeting, not just prediction. DCA shows the model improves screening decisions — 104 additional unmet-need cases per 1,000 screened at threshold 0.5 over "flag everyone," and net-benefit dominance across every threshold a policymaker would consider.

Audit before deployment. The high-income miscalibration would not have surfaced from aggregate Brier alone. Survey-weighted subgroup audits should be standard for targeting models.

Limitations. Unmet need is an access proxy, not a health outcome; temporal robustness on future BRFSS waves is the natural next step.